

Yuze He

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EDUCATION

Carnegie Mellon University

Postdoctoral Fellow in Computer Science Department (CSD)

- Advisors: Peter Steenkiste and Srinivasan Seshan

May 2024 - May 2026

Pittsburgh, USA

The Chinese University of Hong Kong

Ph.D. in Information Engineering (IE)

- Advisor: Guoliang Xing
- Thesis: Leveraging Infrastructure to Enhance Perception and Scene Understanding for Autonomous Driving

Sep 2019 - Dec 2023

Hong Kong, China

Zhejiang University

B.A.Sc. in Electrical Engineering (EE)

- Graduated with highest honors in electrical engineering

Sep 2015 - Jun 2019

Hangzhou, China

HONORS AND AWARDS

- **ACM SIGBED Paul Caspi Memorial Dissertation Award** (2 recipients worldwide) [[🔗](#)] 2026
- **Best Community Contribution Paper Award**, ACM MobiCom 2023 [[🔗](#)] 2023
- **Best Artifact Award**, ACM MobiCom 2024 [[🔗](#)] 2024
- **Best Artifact Award Runner-up**, ACM MobiCom 2024 [[🔗](#)] 2024
- **Best Demo Award Runner-up**, ACM MobiCom 2024 [[🔗](#)] 2024
- **EECS Rising Star**, MIT [[🔗](#)] 2024
- **Doctoral Dissertation Award**, ACM SIGBED China (3 granted annually nationwide) 2024
- **Gold Prize**, 49th International Exhibition of Inventions Geneva 2024
- **Best Paper Award**, The 2023 International Doctoral Forum [[🔗](#)] 2023
- **N2Women Young Researcher Fellowship**, ACM MobiCom 2023 [[🔗](#)] 2023
- **Outstanding Bachelor Dissertation Award**, College of Electrical Engineering, Zhejiang University 2019
- **First Prize**, National Automation System Application Contest, China 2018
- **First Prize** (Rank: 6/230), Undergraduate Electronic Design Contest, Zhejiang Province 2018
- **First Prize** (Rank: 2/67), Undergraduate Smart Car Race, Zhejiang Province 2018

PUBLICATIONS

*Equal contribution (authors listed interchangeably) †Student mentored by Yuze He ‡Co-primary authors

- [1] **Yuze He**, Ferdi Kossmann, Srinivasan Seshan, Peter Steenkiste. "ECCO: Leveraging Cross-Camera Correlations for Efficient Live Video Continuous Learning". In Submission. URL: [🔗](#)
- [2] **Yuze He**^{*}, Jiahe Cui^{*†}, Jianwei Niu, Zhenchao Ouyang, Guoliang Xing. " α LiDAR: An Adaptive High-Resolution Panoramic LiDAR System". In Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (**MobiCom '24**). URL: [🔗](#)
Best Artifact Award (2 of 494 submissions) and **Best Demo Award Runner-up**
Open-source release with **194 GitHub stars** and **6k YouTube demo views** Code: [🔗](#) Demo video: [▶](#)

- [3] **Yuze He**, Chen Bian, Jingfei Xia, Shuyao Shi, Zhenyu Yan, Qun Song, Guoliang Xing. “VI-Map: Infrastructure-Assisted Real-Time HD Mapping for Autonomous Driving”. In Proceedings of the 29th Annual International Conference on Mobile Computing and Networking (**MobiCom '23**). URL: [🔗](#)
Best Community Contribution Paper Award (2 of 377 submissions)
Gold Prize, 49th International Exhibition of Inventions Geneva Code: [🔗](#) Demo video: [🔗](#)
- [4] **Yuze He**, Li Ma, Jiahe Cui, Zhenyu Yan, Guoliang Xing, Sen Wang, Qintao Hu, Chen Pan. “AutoMatch: Leveraging Traffic Cameras to Improve Perception and Localization of Autonomous Vehicles”. In Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems (**SenSys '22**). URL: [🔗](#)
- [5] **Yuze He**, Li Ma, Zhehao Jiang, Yi Tang, Guoliang Xing. “VI-Eye: Semantic-Based 3D Point Cloud Registration for Infrastructure-Assisted Autonomous Driving”. In Proceedings of the 27th Annual International Conference on Mobile Computing and Networking (**MobiCom '21**). URL: [🔗](#)
- [6] Jingfei Xia[†], **Yuze He**, Chen Bian, Zhenyu Yan, Guoliang Xing. “EventEye: Towards High-Frequency Perception Enhancement for Autonomous Vehicles Using Infrastructure-Mounted Event Cameras”. To appear in **SenSys '26**. URL: [🔗](#)
- [7] Neiwen Ling[‡], **Yuze He**[‡], Nan Guan, Heming Fu, Guoliang Xing “An Indoor Smart Traffic Dataset and Data Collection System”. In Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems (**SenSys '22 Workshop Paper**). URL: [🔗](#)
- [8] Jiahe Cui, Shuyao Shi, **Yuze He**, Jianwei Niu, Guoliang Xing, Zhenchao Ouyang. “VILAM: Infrastructure-Assisted 3D Visual Localization and Mapping for Autonomous Driving”. In Proceedings of the 21st USENIX Symposium on Networked Systems Design and Implementation (**NSDI '24**). URL: [🔗](#)
- [9] Shuyao Shi[‡], Neiwen Ling[‡], Zhehao Jiang[‡], Xuan Huang[‡], **Yuze He**, Xiaoguang Zhao, Bufang Yang, Chen Bian, Jingfei Xia, Zhenyu Yan, Raymond W. Yeung, Guoliang Xing. “Soar: Design and Deployment of a Smart Roadside Infrastructure System for Autonomous Driving”. In Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (**MobiCom '24**). **Best Artifact Award Runner-up**. URL: [🔗](#)
- [10] Linlin Tu, Xiaomin Ouyang, Jiayu Zhou, **Yuze He**, Guoliang Xing. “FedDL: Federated learning via dynamic layer sharing for human activity recognition”. In Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems (**SenSys '21**). URL: [🔗](#)
- [11] Neiwen Ling, Kai Wang, **Yuze He**, Guoliang Xing, Daqi Xie. “Rt-mDL: Supporting real-time mixed deep learning tasks on edge platforms”. In Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems (**SenSys '21**). URL: [🔗](#)

PATENTS

- [1] Guoliang Xing, **Yuze He**. “Method for Infrastructure-Assisted Real-Time HD Mapping for Autonomous Driving”. US Application No: 63/586,109

PROPOSAL WRITING EXPERIENCE

NSF VINES Track 2 Proposal

Sep 2025

PI: Vyas Sekar, Anthony Rowe, Srinivasan Seshan, Mayank Goel, Henny Admoni (CMU, Lead); Kittipat Apichart-trisorn, Eugene Chai, TV Lakshman (Nokia Bell Labs); Dave Levin, Amanda Lazar (UMD); Mallesh Dasari (North-eastern); Elahe Soltanaghaei (UIUC).

Role: Contributed to writing one technical thrust and developed visualizations for three use cases.

ACADEMIC SERVICE

Program Committee:

- MobiSys 2025
- IoTDI 2023–2024 (Poster and Demo Tracks)

Artifact Evaluation Committee:

- NSDI 2026
- MobiCom 2024
- MobiSys 2024

Invited Reviewer (Conferences):

- IEEE VR 2026
- CVPR 2024–2026
- ICCV 2025
- ECCV 2025
- ICRA 2024
- INFOCOM 2024

Invited Reviewer (Journals):

- Transactions on Mobile Computing (TMC) 2024–2025
- ACM Transactions on Computing for Healthcare 2024

Organizer and Moderator:

- ACM MobiCom 2023 N2Women Event [\[🔗\]](#)

Event Organizer:

- 17th F1TENTH Autonomous Grand Prix at CPS-IoT Week 2024 [\[🔗\]](#)

Poster Co-Chair:

- Advanced Study Institute 2023 – AI for Internet of Things [\[🔗\]](#)

TEACHING EXPERIENCE

Computer Networks (IERG3310/ESTR3310)	Fall 2021
Graduate Teaching Assistant	CUHK

Computer Networks (IERG3310/ESTR3310)	Spring 2021
Graduate Teaching Assistant	CUHK

Multivariable Calculus for Engineers (ENGG1130)	Spring 2020
Undergraduate Teaching Assistant	CUHK

MENTORING EXPERIENCE

Sruti Srinidhi	Sep 2025 - Present
Ph.D. Student, CMU	

- Mentoring ongoing research on using 2D/3D capture with visual-language models (VLMs) to enable scalable visual search in large indoor environments.

Yoo Chan (Josh) Joung	May 2024 - Aug 2024
M.S. Student, CMU	

- Mentored a summer project on lightweight network telemetry.

Sanjeev Sridhar

M.S. Student, CMU

May 2024 - Aug 2024

- Mentored the same summer project with Josh.

Jingfei Xia

Ph.D. Student, CUHK

Oct 2022 - Jun 2024

- Guided Jingfei's first Ph.D. research project on infrastructure-assisted autonomous driving. Mentored the design, experiments, and paper development of *EventEye* [6], which was later accepted to *SenSys '26*.

Jiahe Cui

Visiting Ph.D. Student, CUHK

June 2022 - May 2024

- Mentored a project on active LiDAR system design, leading to the co-first-authored paper α LiDAR (MobiCom '24) [2], which received both the **Best Artifact Paper Award** and **Best Demo Award Runner-up**.

RESEARCH TALKS

Building Intelligent Infrastructure for Mobile Systems

Oct 2025

CMU Living Edge Lab

Optimizing Streaming of 3D Gaussian Splatting for Large-Scale Virtual Environments

Apr 2025

CMU Living Edge Lab

ECCO: Leveraging Cross-Camera Correlations for Efficient Live Video Continuous Learning

Feb 2025

CMU Living Edge Lab

Infrastructure-Assisted Autonomous Driving: From Perception to HD Mapping

Sep 2024

University of Pittsburgh, Computer Science Department Colloquium

Infrastructure-Assisted Perception and Scene Understanding for Autonomous Driving

Dec 2023

CMU, Hosts: Peter Steenkiste and Srinivasan Seshan

VI-Map: Infrastructure-Assisted Real-Time HD Mapping

Oct 2023

Presented at ACM MobiCom 2023

AutoMatch: Leveraging Traffic Cameras for Vehicle Perception and Localization

Nov 2022

Presented at ACM SenSys 2022

VI-Eye: Semantic 3D Point Cloud Registration between Infrastructure and Vehicles

Mar 2022

Presented at ACM MobiCom 2021 (Online)

REFERENCES

Peter Steenkiste

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Electrical and Computer Engineering

Carnegie Mellon University

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Relationship: Postdoctoral Advisor

Guoliang Xing

Professor in Department of Information
Engineering

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Relationship: Ph.D. Advisor

Chunming Qiao

Professor and Chair of the Computer Science and
Engineering Department

University at Buffalo

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*Relationship: Thesis Committee Member, Academic
Mentor*

Srinivasan Seshan

Professor in Computer Science Department
Carnegie Mellon University

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Relationship: Postdoctoral Advisor

Mahadev Satyanarayanan

Professor in Computer Science Department
Carnegie Mellon University

satya@cs.cmu.edu

Relationship: Academic Mentor